ACTIVITY.

$P_1 = (5,4)$, $P_2 = (7,6)$.

$C_1 = (2,4)$
$C_2 = (4,3)$

at C_1 $(2,4)$ and $(5,4)$.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(5-2)^2 + (4-4)^2} = \sqrt{9} = 3.$$

at $(7,6)$ $d = \sqrt{(7-2)^2 + (6-4)^2} = \sqrt{25+4} = \sqrt{29}$

at C_2 $(4,3)$ and $(5,4)$, $(7,6)$.

$3^2 + 3^2$

$$d = \sqrt{(5-4)^2 + (4-3)^2} = \sqrt{2}$$

$$d = \sqrt{(7-4)^2 + (6-3)^2} = \sqrt{9+9} = \sqrt{18} = 3\sqrt{2}$$

$\therefore$ Since there is no maximum votes, so no votes.

2] $P(8,6)$ and $C(4,3)$.

$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$= \sqrt{(4-8)^2+(3-6)^2} = \sqrt{4^2+3^2} = \sqrt{16+9} = \sqrt{25}$$

$\boxed{\text{radius} = d = 5}$

3] $C(5,4)$ and $P(8,8)$.

$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$= \sqrt{(8-5)^2+(8-4)^2}$$

$$= \sqrt{3^2+4^2} = \sqrt{9+16} = \sqrt{25}$$

$\boxed{d = \text{radius} = 5}$